

PRASANGA DHUNGEL

M.Sc. in Informatics, TUM | Focus: Reliable, Interpretable & Efficient ML

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EDUCATION

Technical University of Munich [*M.Sc. in Informatics, Passed with High Distinction*] **Apr 2023 - Jun 2025**

Thesis: Efficient Large-Scale Data Pruning Using Score Extrapolation **Grade: 1.2/1.0 (Top 10%)**

Supervisor: Prof. Stephan Günnemann **Thesis Grade: 1.0/1.0**

Coursework: Deep Learning, Statistical Learning, ML for Graphs and Sequential Data, Advanced NLP, Deep Generative Models, Computer Vision (Detection, Segmentation, Tracking), Computer Vision (Multiview Geometry), Quantum Computing, Tensor Networks, Deep Learning for Medicine

Institute of Engineering, Pulchowk, Nepal [*B.Sc. in Computer Engineering*] **Nov 2016 - Apr 2021**

Thesis: An Efficient Video Compression Network **Grade: 79.57% (Top 5%)**

Coursework: Theory of Computation, Data Structures and Algorithms, Object Oriented Programming, AI, Computer Architectures, Discrete Structures, Data Mining, Big Data, Calculus, Linear Algebra, Probability, Statistics

PUBLICATIONS

M=Manuscript, C=Conference, †=Equal Contribution

[M1] **Effective Data Pruning through Score Extrapolation**  **2025**

Schmidt Sebastian[†], Dhungel Prasanga[†], et al. *arXiv preprint arXiv:2506.09010*

[C1] **An Efficient Video Compression Network**  **2020**

Dhungel Prasanga, et al. *International Conference on Advances in Computing (IEEE)*

ACHIEVEMENTS AND AWARDS

- ▶ Selected as **top 900 of 50,000** students for **Deutschlandstipendium** (Sponsored by Amazon, €3600) **2024**
- ▶ Selected **top 5 nationally** to represent Nepal at 47th **International Physics Olympiad**, Zürich **2016**
- ▶ Represented Nepal in ACM-ICPC, Dhaka regionals **2018**
- ▶ Ranked **29th of 14,000** applicants nationwide to secure a Full Scholarship for Bachelor's degree **2016**
- ▶ **Hackathons:** Winner of A Day of Code (2020), Locus Codejam (2019), National Level Programming Contest (2018), Second Place at DandyHacks (Rochester University, 2020).

KEY SKILLS

Languages	Python (Expert), C++, C, R, Java, Javascript, MATLAB, Latex, Bash/Shell
AI & ML	PyTorch, TensorFlow, Sklearn, Huggingface, OpenCV, Statsmodel, MLflow, Weights&Biases
MLOps & Cloud	Azure, AWS, Terraform, Docker, CI/CD, Slurm, Airflow, Kubernetes, Prometheus
Data Engineering	PostgreSQL, Elasticsearch, MongoDB, PostGIS, SQLAlchemy, Delta Lake, Spark, Snowflake
Web Development	FastAPI, Django, Flask, REST APIs, Streamlit, Dash, React, Vue.js

WORK EXPERIENCE

E.ON Munich, Germany
Machine Learning Engineer, Full Time **Apr 2025 - Present**

- ▶ Architected end-to-end ML system on **Azure using Terraform** for infrastructure-as-code, enabling reproducible and scalable deployments with **99% uptime**.
- ▶ Built a comprehensive model lifecycle management using **MLflow** for experiment tracking, hyperparameter optimization, versioning, and performance monitoring.
- ▶ Led recruitment, onboarding, and mentorship of two junior data scientists, fostering a high-performing, collaborative, and learning-focused environment.

Data Scientist, Working Student **Sep 2023 - Mar 2025**

- ▶ Developed novel **Robust-PLS algorithm** optimized for high-dimensional, correlated feature spaces with outliers, achieving **7% improvement in MAE**.
- ▶ Implemented custom noise detection and reduction algorithms, improving sensor reliability metrics across deployment scenarios.

- ▶ Developed end-to-end ML systems for real estate investment optimization, processing \$20M+ in property evaluations. Trained **ensemble of ARIMA, Prophet, and LSTM** to forecast revenue that only had 14% of error on hindsight. Worked on a physical and geospatial feature extraction pipeline, analyzing 100K+ properties.
- ▶ Architected a **conversational real estate assistant**. Implemented BERT-based intent classification and developed custom NLG (GPT-2 fine-tuning) for human-like property descriptions.

PROJECTS AND RESEARCH

Effective Data Pruning through Score Extrapolation, *Master's Thesis* 2025

We proposed a novel framework for large-scale dataset pruning that eliminates costly full-dataset scoring. Our method extrapolates importance scores using geometric data structure (KNN, GNNs), yielding competitive model performance with substantial end-to-end training time savings across multiple benchmarks and pruning techniques.

Video Compression Net, *Bachelor's Thesis* 2020

We introduced a Deep Learning based video compression system that minimizes the number of bits required to encode video frames while also reducing distortion. The major objective of the project was to replace the conventional components in the pipeline of video compression with their neural network equivalents. The trained model equaled the compression ratio of the conventional system.

LANGUAGE

English: Full Professional Proficiency (TOEFL 109) | **German:** Basic Conversational | **Nepali:** Mother tongue